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Internal combustion engine

Abstract

An internal combustion engine includes a crankcase for rotatably supporting a crankshaft about a crankshaft longitudinal axis. The crankcase comprises a crankcase water jacket, a cylinder head which is connected to the crankcase and comprises a cylinder head water jacket, and a coolant pump for supplying the crankcase water jacket and the cylinder head water jacket with a coolant volume flow. The crankcase water jacket and the cylinder head water jacket are fluidly connected to one another in series. The crankcase water jacket is suppliable with the coolant volume flow by the coolant pump via a crankcase inlet, and the crankcase water jacket is designed such that the coolant volume flow is directed via a crankcase outlet and is transferred in the direction of the cylinder head water jacket. The crankcase inlet and the crankcase outlet are axially spaced apart. The crankcase inlet is arranged on a first longitudinal side of the crankcase.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4212270	12/1979	Nakanishi	123/41.46	F01P 7/165
9140176	12/2014	Steiner	N/A	F01P 5/10
9500115	12/2015	Quix	N/A	F01P 7/165
10934926	12/2020	Steiner	N/A	F01M 1/16
2002/0189558	12/2001	Batzill et al.	N/A	N/A
2003/0111025	12/2002	Kim	123/41.72	F02F 1/14
2015/0369167	12/2014	Beyer	123/41.74	F02F 1/10
2016/0010533	12/2015	Matsumoto	123/41.08	F02F 1/14
2017/0306831	12/2016	Craft	N/A	F01P 3/14
2018/0347443	12/2017	Boehmer	N/A	F02F 1/36
2020/0232412	12/2019	Watanabe et al.	N/A	N/A
2021/0317770	12/2020	Tada	N/A	F01P 3/02

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102015014514	12/2016	DE	F01P 3/02
102017003310	12/2016	DE	F01P 3/02
2017125445	12/2016	JP	N/A
20070040217	12/2006	KR	N/A
WO0183959	12/2000	WO	N/A
WO0188359	12/2000	WO	N/A

OTHER PUBLICATIONS

DE-102017003310-a1 (Year: 2017). cited by examiner

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Background/Summary

(1) This claims the benefit of German Patent Application DE 10 2022 003 904.4, filed on Oct. 13, 2022 which is hereby incorporated by reference herein.

(2) The present disclosure relates to an internal combustion engine having a crankcase water jacket and a cylinder head water jacket which are fluidly connected to each other.

BACKGROUND

(3) Internal combustion engines are subject to high thermal stress in the area of the crankcase and cylinder head. To reduce this thermal stress, modern internal combustion engines are water-cooled. For this purpose, internal combustion engines can comprise a water jacket with a crankcase water jacket and a cylinder head water jacket. The water jacket is regularly integrated into the crankcase or cylinder head during the casting process by inserting casting cores.

(4) JP 2017-125445 A discloses an internal combustion engine having a crankcase water jacket. The crankcase water jacket is supplied with a coolant volume flow by a coolant pump via a crankcase inlet, which is arranged on one front side of the crankcase.

(5) DE 10 2015 014 514 A1 discloses an internal combustion engine with a water jacket comprising a crankcase water jacket and a cylinder head water jacket, which are fluidly connected in series. The crankcase water jacket is fed by an inlet rail with a coolant volume flow. The coolant flow is distributed to multiple outlets in the crankcase water jacket. At each one of these multiple outlets, a portion of the coolant volume flow is transferred directly through the cylinder head gasket to the cylinder head water jacket. This arrangement shows the disadvantage that the design of the transfer from crankcase water jacket to cylinder head water jacket is installation space intensive and prevents additional integration of functional sections into the water jacket.

SUMMARY

(6) Starting from this, an objective of the present disclosure is providing an internal combustion engine that comprises a water jacket optimized for installation space and enables the integration of further functional sections into the water jacket.

(7) To solve a problem underlying the present disclosure, an internal combustion engine is provided comprising: a crankcase for rotatably supporting a crankshaft about a crankshaft longitudinal axis, the crankcase comprising a crankcase water jacket, a cylinder head connected to the crankcase and comprising a cylinder head water jacket, and a coolant pump for supplying the crankcase water jacket and the cylinder head water jacket with a coolant volume flow, the crankcase water jacket and the cylinder head water jacket being fluidly connected to one another in series, wherein the crankcase water jacket is suppliable with an entire coolant volume flow from the coolant pump via a crankcase inlet, and the crankcase water jacket is configured such that the entire coolant volume flow is drained via a crankcase outlet and transferred in the direction of the cylinder head water jacket, and the crankcase inlet and the crankcase outlet are axially spaced apart from one another, wherein the crankcase inlet is arranged on a first longitudinal side of the crankcase.

(8) In the internal combustion engine according to the present disclosure, the coolant volume flow provided by the coolant pump is thus completely drained or discharged from the crankcase water jacket through the crankcase outlet, wherein the complete coolant volume flow has previously flowed through the crankcase water jacket at least in the longitudinal direction. The resulting single outlet of the crankcase water jacket creates free installation space in the internal combustion engine which can be used, for example, to integrate further functional sections of the water jacket.

(9) In one possible embodiment of the internal combustion engine, the crankcase outlet may be arranged on a second longitudinal side of the crankcase. In this way, in addition to the flow through the crankcase water jacket in the longitudinal direction, an additional flow through the crankcase water jacket by the complete coolant volume flow in the transverse direction can be realized. The resulting complete longitudinal-transverse flow of the coolant through the crankcase allows it to be cooled efficiently.

(10) In another possible embodiment of the internal combustion engine, the crankcase may

comprise a first cylinder bore and a second cylinder bore separated from each other in the axial direction by a cylinder web section. The crankcase inlet can be arranged in the axial direction on a first side of the cylinder web section, and the crankcase outlet can be arranged on a second side of the cylinder web section opposite the first side.

(11) In another possible embodiment of the internal combustion engine, the crankcase may include a line of cylinder bores, the first cylinder bore delimiting the line of cylinder bores as a first cylinder end bore and the second cylinder bore delimiting the series of cylinder bores as a second cylinder end bore. The first cylinder bore and the second cylinder bore can each be arranged in the axial direction, i.e. in a direction parallel to the longitudinal axis of the crankshaft, between the crankcase inlet and the crankcase outlet, at least in sections.

(12) In another possible embodiment of the internal combustion engine, the cylinder head water jacket can be supplied with at least part of the coolant volume flow by the coolant pump via the crankcase water jacket and a cylinder head inlet. The coolant volume flow can be transferred to the coolant pump via a coolant pump inlet. The cylinder head inlet can be located on the second longitudinal side and the coolant pump inlet can be located on the first longitudinal side of the crankcase.

(13) The cylinder head inlet can also be spaced apart from the coolant pump inlet in the axial direction. The crankcase inlet can be arranged in the axial direction between the coolant pump inlet and the crankcase outlet. The first cylinder bore and the second cylinder bore can each be arranged in the axial direction between the crankcase outlet and the cylinder head inlet, at least in sections.

(14) In one possible embodiment of the internal combustion engine, the crankcase outlet and the cylinder head inlet can be fluidly connected to each other via a longitudinal channel arranged in the crankcase. An oil heat exchanger can be arranged in the longitudinal channel in an oil heat exchanger recess.

(15) In another possible embodiment of the internal combustion engine, the crankcase inlet may open into a crankcase distributor channel. The crankcase distributor channel may be arranged on the first longitudinal side of the crankcase. A crankcase collecting channel disposed on the second longitudinal side of the crankcase may open into the crankcase outlet. The crankcase distributor channel and the crankcase collecting channel can be fluidly connected to each other via at least one cylinder web channel. A width of the crankcase distributor channel may decrease in the axial direction away from the crankcase inlet. A width of the crankcase collecting channel may increase in the axial direction toward the crankcase outlet.

(16) In another possible embodiment of the internal combustion engine, the cylinder head inlet may open into a cylinder head distributor channel. The cylinder head distributor channel can be arranged on the second longitudinal side of the crankcase. A cylinder head collecting channel located on the first longitudinal side of the crankcase can open into the coolant pump inlet. The cylinder head distributor channel and the cylinder head collecting channel can be fluidly connected to each other via at least one cylinder head transverse channel.

(17) A width of the cylinder head distributor channel can decrease in the axial direction away from the cylinder head inlet. A width of the cylinder head collecting channel can increase in the axial direction towards the coolant pump inlet.

(18) The crankcase distributor channel can be located between the cylinder head collecting channel and a plane defined by a cylinder head gasket in the crankcase. The crankcase collecting channel can be arranged in the crankcase between the longitudinal channel and a plane defined by a cylinder head gasket.

Description

BRIEF SUMMARY OF THE DRAWINGS

- (1) A possible embodiment of the internal combustion engine according to the present disclosure is explained below with reference to the figures. Herein,
- (2) FIG. 1 shows a first perspective view of the water jacket of an internal combustion engine according to the present disclosure;
- (3) FIG. 2 shows a second perspective view of the water jacket of the internal combustion engine from FIG. 1;
- (4) FIG. 3 shows a perspective view of the crankcase water jacket of the internal combustion engine of FIG. 1 together with the longitudinal channel,
- (5) FIG. 4 shows a top view of the crankcase water jacket from FIG. 3;
- (6) FIG. 5 shows a perspective view of the cylinder head distributor channel and the cylinder head collecting channel, and
- (7) FIG. 6 shows a top view of the cylinder head distributor channel and the cylinder head collecting channel from FIG. 5.

DETAILED DESCRIPTION

- (8) FIGS. 1 to 6, which are described together below, show the water jacket 1 of an internal combustion engine according to the present disclosure. In the figures, the water jacket 1 is represented in the manner known to those skilled in the art by the cast cores used. The internal combustion engine according to the present disclosure has a longitudinal axis L_{long} , which is defined by the crankshaft axis of the internal combustion engine. In the present case, the internal combustion engine is designed as an in-line four-cylinder with a first cylinder bore 12, a second cylinder bore 15, a third cylinder bore 13 and a fourth cylinder bore 14. The first cylinder bore 12 delimits the cylinder line on a first side and may therefore be referred to as the first cylinder end bore. The second cylinder bore 15 delimits the cylinder line on a second side and may therefore be referred to as a second cylinder end bore. The cylinder bores 12, 13, 14, 15 each have parallel cylinder bore axes.
- (9) In FIG. 4, cylinder casting cores are indicated as dashed circles in the respective cylinder bores. It can thus be seen that the first cylinder bore 12 and the third cylinder bore 13 are separated from each other by a first cylinder web section 26, the third cylinder bore 13 and the fourth cylinder bore 14 are separated from each other by a second cylinder web section 27, and the fourth cylinder bore 14 and the second cylinder bore 15 are separated from each other by a third cylinder web section 28.
- (10) The water jacket 1 includes a crankcase water jacket 5 disposed in the crankcase of the internal combustion engine, and a cylinder head water jacket 18 disposed in the cylinder head of the internal combustion engine. The crankcase and the cylinder head of the internal combustion engine are sealed against each other in a known manner by a cylinder head gasket, which is not shown. In this regard, the cylinder head gasket includes passages that allow fluidic communication between the crankcase water jacket 5 and the cylinder head water jacket 18.
- (11) The water jacket 1 of the internal combustion engine represents a circuit which can be flown through by a coolant pump 31. Herein, the crankcase water jacket 5 and the cylinder head water jacket 18 are arranged in series. Starting from the coolant pump 31, the coolant in the direction of flow first flows through the crankcase water jacket 5 before subsequently flowing through the cylinder head water jacket 18. The flow path of the coolant is undoubtedly apparent to the skilled person from FIGS. 1 to 6, the flow of the coolant being indicated by arrows as an supplement. For clarity, the flow of coolant between the coolant pump 31 and the initial entry into the cylinder head water jacket 18 is shown with dashed-dotted arrows, and between the initial entry of coolant into the cylinder head water jacket 18 to the re-entry of coolant into the coolant pump 31 is shown with dashed arrows.
- (12) The crankcase water jacket 5 is formed by a crankcase main core 29. A core for a longitudinal

channel **16**, a core for a cylinder head distributor channel **20** and a core for a cylinder head collecting channel **23** are also arranged in the crankcase.

(13) The cylinder head water jacket **18** is formed by a cylinder head main core **30**. The cylinder head main core **30** comprises a section **2** for accommodating the coolant pump **31**, a thermostat channel section **3** in which a thermostat can be arranged to control the coolant flow and which forms the inlet to an external cooler, and a suction channel section **4** which supplies the suction side of the coolant pump **31** with coolant and is fluidly connected to the external cooler. In principle, it is also conceivable that the section **2**, the thermostat channel section **3** and the suction channel section **4** are designed in a separate core, which is arranged in the crankcase or a separate bracket, for example.

(14) The crankcase main core **29** has a crankcase inlet **6** which is fluidly connected to the pressure side of the coolant pump **31** via a passage in the cylinder head gasket. The crankcase inlet **6** opens into a crankcase distributor channel **7**. The crankcase distributor channel **7** tapers at least in sections starting from the crankcase inlet **6** along the longitudinal axis L_{long} of the internal combustion engine. The crankcase main core **29** also has a crankcase collecting channel **10**, which is fluidly connected to the crankcase distributor channel **7** via three web channels **8** as well as a first front channel **9** and a second front channel **9'**. A first web channel **8** extends through the first cylinder web section **26**, a second web channel **8'** extends through the second cylinder web section **27**, and a third web channel **8''** extends through the third cylinder web section **28**. The first front channel **9** extends through a first front wall of the crankcase and the second front channel extends through a second front wall of the crankcase.

(15) The crankcase collecting channel **10** opens into a crankcase outlet **11** and widens at least in sections along the longitudinal axis L_{long} towards the crankcase outlet **11**. In the present case, the crankcase water jacket **5** represents a closed system or closed circuit in the area between the crankcase inlet **6** and the crankcase outlet **11**. Thus, an entire flow delivered by the coolant pump **31** to the crankcase inlet **6** is completely directed via the crankcase outlet **11**.

(16) The crankcase inlet **6** and the crankcase outlet **11** are spaced from each other along the longitudinal axis L_{long} . The cylinder bores **12**, **13**, **14**, **15** of the internal combustion engine are arranged at least partially between the crankcase inlet **6** and the crankcase outlet **11** in the direction of the longitudinal axis L_{long} . In the present case, the third cylinder bore **13** and the fourth cylinder bore **14** are arranged entirely between the crankcase inlet **6** and the crankcase outlet **11**. With respect to the longitudinal axis L_{long} , the first cylinder bore **12** overlaps radially with the crankcase inlet **6**, at least in sections. With respect to the longitudinal axis L_{long} , the second cylinder bore **15** overlaps radially with the crankcase outlet **11**, at least in sections. This arrangement of crankcase inlet **6** and crankcase outlet **11** can ensure that the coolant flow passes completely through the crankcase in the longitudinal direction.

(17) The crankcase inlet **6** is arranged on the first longitudinal side of the crankcase, on which the coolant pump **31** is also arranged. The crankcase outlet **11** is arranged on the second longitudinal side of the crankcase, which is opposite the first longitudinal side. This arrangement of crankcase inlet **6** and crankcase outlet **11** can ensure that the crankcase is flowed through by the entire coolant volume flow in a direction transverse to the longitudinal axis L_{long} . The crankcase is thus flown through longitudinally-transversely by the entire coolant volume flow.

(18) The crankcase outlet **11** opens into a longitudinal channel **16** cast into the crankcase. The longitudinal channel **16** extends from the crankcase outlet **11** in a direction parallel to the longitudinal axis L_{long} towards a cylinder head inlet **19**. In the present case, the cylinder head inlet **19** is arranged in radial overlap with the first cylinder bore **12** with respect to the longitudinal axis L_{long} .

(19) Between the crankcase outlet **11** and the cylinder head inlet **19**, the longitudinal channel **16** comprises a heat exchanger recess **17** in which a heat exchanger, for example a heat exchanger for an oil cooling circuit of the internal combustion engine, can be arranged.

(20) The cylinder head inlet **19** opens into the cylinder head distributor channel **20**. The cylinder head distributor channel **20** is cast in the crankcase. At least part of the coolant flow provided by the coolant pump **31** flows through the cylinder head inlet **19**. In particular, the complete coolant flow provided by the coolant pump **31** can flow through the cylinder head inlet **19**. The cylinder head distributor channel **20** extends in a direction parallel to the longitudinal axis L_{long} and ends in axial direction in before of the crankcase outlet **11**. Thus, the crankcase outlet **11** and the cylinder head distributor channel **20** do not interpenetrate. The cylinder head distributor channel **20** is arranged on the second longitudinal side of the crankcase respectively cast into it. The cylinder head distributor channel **20** has a central section that extends parallel to the longitudinal axis L_{long} and tapers in the direction of the longitudinal axis L_{long} away from the cylinder head inlet **19**. In the present case, the width of the central section of the cylinder head distributor channel **20** and the width of the crankcase collecting channel **10** change in opposite directions in a direction parallel to the longitudinal axis L_{long} .

(21) In the present case, the cylinder head distributor channel **20** has four transfer sections **21** extending substantially parallel to the cylinder bore axes. The four transfer sections **21** are evenly spaced apart from each other in the direction of the longitudinal axis L_{long} . The four transfer sections **21** each end in a transfer opening **31** which cooperate with passages in the cylinder head gasket so that coolant can be transferred from the respective transfer section **21** into the cylinder head water jacket **18**.

(22) The cylinder head water jacket **18** includes a plurality of cylinder head transverse channel **22**, which are fed with coolant from the transfer sections **21** and extend transversely to the longitudinal axis L_{long} . The cylinder head water jacket **18** is designed in such a way that the coolant flows through the cylinder head essentially in a direction transverse to the longitudinal axis L_{long} .

(23) A cylinder head collecting channel **23** is arranged on respectively cast into the first longitudinal side of the crankcase. The cylinder head collecting channel **23** comprises four takeover sections **24** extending substantially parallel to the cylinder bore axes and opening into a central section of the cylinder head collecting channel **23**, wherein the central section extends in the direction of the longitudinal axis L_{long} of the internal combustion engine. The central section of the cylinder head collecting channel **23** widens in the direction of the longitudinal axis L_{long} toward the coolant pump inlet **25**. The width of the central section of the cylinder head collecting channel **23** and the width of the crankcase distributor channel **7**, in the present case, change in the same direction parallel to the longitudinal axis L_{long} .

(24) The takeover sections **24** are uniformly spaced apart from each other in the direction of the longitudinal axis L_{long} . The takeover sections **24** each terminate in a transfer opening **32** which cooperates with passages in the cylinder head gasket so that coolant can be transferred from the cylinder head water jacket **18** into the respective transfer section **24**. The central section of the cylinder head collecting channel **23** opens into a coolant pump inlet **25**, through which the coolant is returned past the thermostat and external cooler to the coolant pump **31**.

(25) The cylinder head inlet **19** is arranged between the coolant pump inlet **25** and the crankcase outlet **11** in a direction parallel to the longitudinal axis L_{long} .

(26) The crankcase distributor channel **7** is arranged in the crankcase and in a direction parallel to the cylinder bore axes at least in sections between the cylinder head collecting channel **23** and a plane defined by a cylinder head gasket. In the present case, the crankcase distributor channel **7** is arranged between the central portion of the cylinder head collecting channel **23** and the plane defined by a cylinder head gasket in the crankcase. As can be seen in particular from FIG. 2, the crankcase distributor channel **7** has recesses for this purpose through which the takeover sections **24** of the cylinder head collecting channel **23** extend.

(27) The crankcase collecting channel **10** is arranged in the crankcase and in a direction parallel to the cylinder bore axes at least in sections between the longitudinal channel **16** and the plane defined by the cylinder head gasket.

LIST OF REFERENCE SIGNS

(28) **1** Water jacket **2** Section to accommodate a coolant pump **3** Thermostat channel section **4** Suction channel section **5** Crankcase water jacket **6** Crankcase inlet **7** Crankcase distributor channel **8** Web channel **9** Front channel **10** Crankcase collecting channel **11** Crankcase outlet **12** Cylinder bore **13** Cylinder bore **14** Cylinder bore **15** Cylinder bore **16** Longitudinal channel **17** Heat exchanger recess **18** Cylinder head water jacket **19** Cylinder head inlet **20** Cylinder head distributor channel **21** Transfer section **22** Cylinder head transverse channel **23** Cylinder head collecting channel **24** Takeover section **25** Coolant pump inlet **26** Cylinder web section **27** Cylinder web section **28** Cylinder web section **29** Crankcase main core **30** Cylinder head main core **31** Coolant pump L_long Crankshaft axle

Claims

1. An internal combustion engine comprising: a crankcase for rotatably supporting a crankshaft about a crankshaft longitudinal axis, the crankcase comprising a crankcase water jacket; a cylinder head connected to the crankcase and comprising a cylinder head water jacket; and a coolant pump for supplying the crankcase water jacket and the cylinder head water jacket with a coolant volume flow, the crankcase water jacket and the cylinder head water jacket being fluidly connected to one another in series, the crankcase water jacket being suppliable with an entire coolant volume flow by the coolant pump via a single crankcase inlet, the crankcase water jacket being designed such that the entire coolant volume flow is directed via a single crankcase outlet and is transferred in the direction towards the cylinder head water jacket, the crankcase inlet and the crankcase outlet being axially spaced apart, the crankcase inlet being arranged on a first longitudinal side of the crankcase, wherein the crankcase comprises a first cylinder bore and a second cylinder bore which are separated from one another in the axial direction by a cylinder web section, wherein the crankcase inlet is arranged in the axial direction on a first side of the cylinder web section and the crankcase outlet is arranged on a second side of the cylinder web section opposite the first side, wherein the crankcase comprises a line of cylinder bores, wherein the first cylinder bore delimits the line of cylinder bores as a first cylinder end bore and the second cylinder bore delimits the line of cylinder bores as a second cylinder end bore, wherein the first cylinder bore and the second cylinder bore are each arranged in the axial direction between the crankcase inlet and the crankcase outlet, at least in sections, wherein the cylinder head water jacket is supplied with at least part of the coolant volume flow by the coolant pump via the crankcase water jacket and a cylinder head inlet, and wherein the coolant volume flow is transferred from the cylinder head water jacket via a coolant pump inlet to the coolant pump, wherein the cylinder head inlet is arranged on the second longitudinal side and the coolant pump inlet is arranged on the first longitudinal side of the crankcase.

2. The internal combustion engine according to claim 1, wherein the crankcase outlet is arranged on a second longitudinal side of the crankcase.

3. The internal combustion engine according to claim 1, wherein the cylinder head inlet is arranged at a distance in the axial direction from the coolant pump inlet.

4. The internal combustion engine according to claim 1, wherein the first cylinder bore and the second cylinder bore are each arranged in the axial direction between the crankcase outlet and the cylinder head inlet, at least in sections.

5. The internal combustion engine according to claim 1, wherein the crankcase inlet is arranged in the axial direction between the coolant pump inlet and the crankcase outlet.

6. The internal combustion engine according to claim 1, wherein the crankcase outlet and the cylinder head inlet are fluidly connected via a longitudinal channel in the crankcase, wherein an oil heat exchanger is arranged in an oil heat exchanger recess in the longitudinal channel, in particular.

7. The internal combustion engine according to claim 2, wherein the crankcase inlet opens into a

crankcase distributor channel which is arranged on the first longitudinal side of the crankcase; and wherein a crankcase collecting channel, which is arranged on the second longitudinal side of the crankcase, opens into the crankcase outlet; wherein the crankcase distributor channel and the crankcase collecting channel are fluidly connected to one another via at least one cylinder web channel.

8. The internal combustion engine according to claim 7, wherein a width of the crankcase distributor channel decreases in axial direction away from the crankcase inlet, and wherein a width of the crankcase collecting channel increases in the axial direction toward the crankcase outlet.

9. An internal combustion engine comprising: a crankcase for rotatably supporting a crankshaft about a crankshaft longitudinal axis, the crankcase comprising a crankcase water jacket; a cylinder head connected to the crankcase and comprising a cylinder head water jacket; and a coolant pump for supplying the crankcase water jacket and the cylinder head water jacket with a coolant volume flow, the crankcase water jacket and the cylinder head water jacket being fluidly connected to one another in series, the crankcase water jacket being suppliable with an entire coolant volume flow by the coolant pump via a single crankcase inlet, the crankcase water jacket being designed such that the entire coolant volume flow is directed via a single crankcase outlet and is transferred in the direction towards the cylinder head water jacket, and the crankcase inlet and the crankcase outlet being axially spaced apart, the crankcase inlet being arranged on a first longitudinal side of the crankcase, wherein the crankcase outlet is arranged on a second longitudinal side of the crankcase; wherein the cylinder head inlet opens into a cylinder head distributor channel which is arranged on the second longitudinal side of the crankcase; wherein a cylinder head collecting channel, which is arranged on the first longitudinal side of the crankcase, opens into the coolant pump inlet; and wherein the cylinder head distributor channel and the cylinder head collecting channel are fluidly connected to one another via at least one cylinder head transverse channel.

10. The internal combustion engine according to claim 9, wherein a width of the cylinder head distributor channel decreases in the axial direction away from the cylinder head inlet, and/or wherein a width of the cylinder head collecting channel increases in the axial direction toward the coolant pump inlet.

11. The internal combustion engine according to claim 9, wherein the crankcase distributor channel is arranged in the crankcase between the cylinder head collecting channel and a plane defined by a cylinder head gasket.

12. The internal combustion engine according to claim 11, wherein the crankcase collecting channel is arranged in the crankcase between the longitudinal channel and the plane defined by the cylinder head gasket.
